DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter 23

Course: B. Tech.Branch : Computer Engineering and Allied **Semester : VI**

Subject Code & Name: BTCOC601_Y20 -Compiler Design

Max Marks: 60 **Date: 16/01/2024** **Duration: 3 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	Draw a neat diagram and explain different phases of compiler.	Remembering	**6**
B)	What do you understand by "Input Buffering"? Explain 'Buffer pairs' and sentinels.	Remembering	**6**
C)	List out and explain different compile-construction tools.	Analyzing	**6**
Q.2	**Solve Any Two of the following.**		**12**
A)	What is a finite automata? Construct transition diagram of a DFA for the string $(a+b)^*$.	Applying	**6**
B)	What is Lex? Explain.	Applying	**6**
C)	Give the formal definition of Context-Free Grammar.	Analyzing	**6**
Q.3	**Solve Any Two of the following.**		**12**
A)	What is Parse Tree? Explain with example.	Remembering	**6**
B)	What is LL(1) Grammar?	Remembering	**6**
C)	Explain Bottom-Up parsing.	Applying	**6**
Q.4	**Solve Any Two of the following.**		**12**
A)	What is Syntax-Directed Definition?	Understanding	**6**
B)	Explain type checking in detail.	Analyzing	**6**
C)	What is Symbol Table? Explain the organization of symbol table in detail.	Applying	**6**
Q.5	**Solve Any Two of the following.**		
A)	Define the DAG representation of basic blocks.	Applying	**6**
B)	Explain the various storage allocation strategies?	Applying	**6**
C)	What is syntax directed translation in compiler design?	Undestanding	**6**

***** End *****

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